

a

Neutral-drift ladder (brain, library-level pairs)

T1 technical drift

same donor, same region, two 10x libraries
ground truth: no functional difference

T2 donor drift

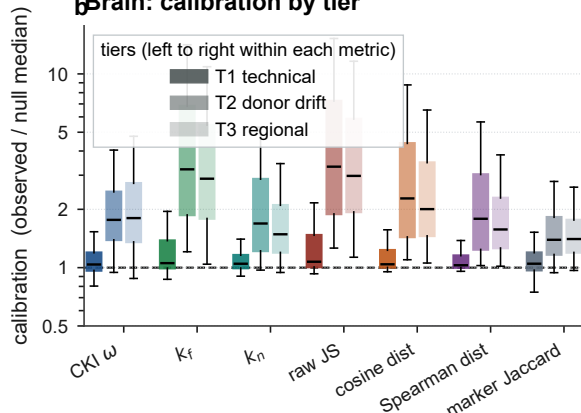
same region, two donors
technical + inter-individual drift

T3 regional biology

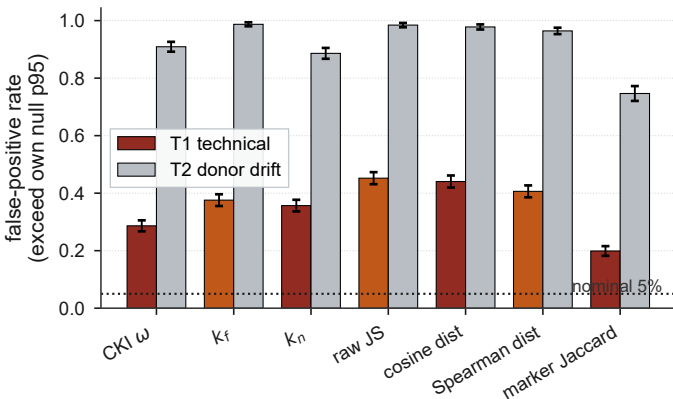
same donor, two regions
positive control: divergence expected

per pair: observed vs. n -matched cell-shuffle null
(pooled nuclei, split at observed group sizes, $B = 100/30/30$)

Brain: calibration by tier



Brain: drift misreporting



Replication: Kang batch1 technical replicates

